CAIEMS Pipeline: A Real Accident Narrative Processed End-to-End

Raw Accident Narrative

Chengdu, Jinjiang District: a temporary scaffold at the Shahe Yihao residential community partially collapsed, injuring one person. According to a June 16 briefing, the scaffold suddenly collapsed around 11 a.m., trapping one person. The trapped person was rescued by multiple agencies and taken to hospital with stable vital signs.

CAICS
Knowledge Base
5 layers · 19 causal categories
M→F→T→H→C

MECE Enumerations
5 fields
work task 11 · equipment 13
material 7 · worker 13
structure 12

CAICS Operational Definitions (excerpt)

M1 Deficient site supervision: safety supervision did not cover critical work faces; unsafe conditions were not detected and stopped before the accident.
F2 Unstable support systems: formwork, scaffolding, or pit support failed due to insufficient capacity or structural defects.
H1 Procedure violation: acts explicitly violating work plans, safety procedures, or technical codes.

Definitions + **Accident narrative** + **Instruction**

Stage-1: Text Classification & Basic Field Extraction

33 structured fields via tool_use

- Classification: single (one accident report)
- accident_type = Collapse
- severity = General accident
- deaths = 0 / injuries = 1
- city = Chengdu
- work_task = Decoration & renovation works

weather / project participants / construction phase + 27 other fields…

Output: 33-field structured record (schema-enforced output)

LLM
DeepSeek V4

Stage-2: CAICS Causal Coding

Two-step judgment + evidence chains

- M1 Deficient site supervision
- M2 Failed hazard inspection
- M3 Insufficient safety training
- M4 Procedural gaps
- F2 Unstable support systems
- T2 Abnormal material state
- H1 Procedure violation
- H3 Failed risk perception

Each code carries: verbatim evidence + inference chain + explicit/inferred label

Final Analysis Report — Database Record

accidents table id = fffab71e9e4d | title = Partial scaffold collapse at Shahe Yihao residential community, Chengdu
Basic fields type = Collapse · severity = General · deaths = 0 · injuries = 1 · city = Chengdu
CAICS codes [M1, M2, M3, M4, F2, T2, H1, H3] ← 8 causes, each with verbatim evidence (auditable)
Enum standardization equipment = EQP05 Scaffolding · worker = WKR12 External personnel
Enrichment search-enhanced news · CLIP-filtered image · VLM description (80–150 words)

The processed record joins **2,623 accident records** and **17,364 evidence-bound causal codes**, retrievable via the Web platform and REST API